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Green House Effect

The presence of atmosphere in the earth plays very important role in human lives. Top of the atmosphere is at -19°C and bottom of the atmosphere is at +14°C. The increase in 33°C from top to bottom is due to some gases present in the atmosphere. These gases are called Greenhouse gases and this effect is called Greenhouse effect.

The greenhouse gases are mainly CO_2, water vapour, Ne, He, NO_2, CH_4, Xe, Kr, ozone and NH_3. Except CO_2 and water vapor, all others are present only in very small amount in the atmosphere. The radiation from the Sun is mainly in the visible region of the spectrum. The earth absorbs these radiations and reradiate in the infrared region. Carbon dioxide and water Vapour are good absorbers of infrared radiation since they have more vibrational degree of freedom compared to nitrogen and oxygen (you will learn in unit 9) which keeps earth warmer as shown in Figure.

The amount of CO_2 present in the atmosphere is increased from 20% to 40% due to human activities since 1900s. The major emission of CO_2 comes from burning of fossil fuels. The increase in automobile usage worldwide causes this damage. Due to this increase in the CO_2 content in the atmosphere, the average temperature of the earth increases by 1°C. This effect is called global warming. It has serious influence and alarming effect on ice glaciers on Arctic and Antarctic regions. In addition, the CO_2 content is also increasing in ocean which is very dangerous to species in the oceans.

In addition to CO_2, another very important greenhouse gas is Chloro flouro carbon(CFC) which is used as coolant in refrigerators worldwide. In the human made greenhouse gases CO_2 is 55%, CFCs are 24%. Nitrogen oxide is 6% and methane is 15%. CFCs also has made huge damage to ozone layer.

Lot of efforts are taken internationally to reduce the emission of CO_2 and CFCs in various countries. Nowadays a lot of research is going to replace non fossil fuels to replace the fossil-fuels in automobile industry. The major emission of CO_2 comes from developed countries like USA and European countries. Various treaties are formed between countries to reduce the emission of CO_2 to considerable level before 2020s. But still global warming is not taken seriously in various countries.

